

BEC305 Python Programming — syllabus extract

Extracted verbatim from 3ecsy11.pdf (238,451 characters of machine-readable text).

the students'
understanding.

14

Python Programming

Course Code

1BEC305

Scheme

2025

Type of Course

PCC

Semester

III

Teaching Hours/Week (L:T:P)

3:0:0

CIE Marks

50

Total Hours of Pedagogy (

L

:T:P:SL&TW)

42:0:0:48

SEE Marks

50

Credits

3

Total Marks

100

Examination type (SEE)

Theory

Exam Hours

3

Course outcome (Course Skill Set) At the end of the course, the student will be able to:

1.

Acquire programming skills in Python

2.

Demonstrate data structure representation using Python

3.

Develop the skill of pattern matching and files in Python

4.

Acquire Object Oriented Skills in Python

5.

Develop the ability to write database applications in Python

Module-1

Python Basics, Python language features, History, Entering Expressions into the Interactive Shell, The Integer, Floating-Point, and String Data Types, String Concatenation and Replication, Storing Values in Variables, Your First Program, Dissecting Your Program, Flow control, Boolean Values, Comparison Operators, Boolean Operators, Mixing Boolean and Comparison Operators, Elements of Flow Control, Program Execution, Flow Control Statements, Importing Modules, Ending a Program Early with `sys.exit()`, Functions, def Statements with Parameters, Return Values and return Statements, The None Value, Keyword Arguments and `print()`, Local and Global Scope, The global Statement, Exception Handling, A Short Program: Guess the Number

Text book 1: Chapters 1

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3

Number of Hours:9

Module-2

Data Structures: Lists: The List Data Type, Working with Lists Strings: Manipulating Strings, Working with Strings, Useful String Methods Tuples and Dictionaries, basics Using Data Structures to Model Real-World Things, Manipulating Strings.

Text book 1: Chapters 4

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6

Number of Hours:8

Module-3

Pattern Matching with Regular Expressions, Finding Patterns of Text Without Regular Expressions, Finding Patterns of Text with Regular Expressions, More Pattern Matching with Regular Expressions,, The `findall()` Method, Character Classes, Making Your Own Character Classes, The Caret and Dollar Sign Characters, The Wildcard Character, Review of Regex Symbols. Reading and Writing Files, Files and File Paths, The `os.path` Module, The File Reading/Writing Process, Saving Variables with the `shelve` Module, Saving Variables with the `pprint`. `pformat()` Function

Text book 1: Chapters 7, 8

Numbe

r of Hours:8

Module-4

Classes and objects: Programmer-defined types, Attributes, Rectangles, Instance

es as return values, Objects are mutable, Copying, Classes and functions: Time, Pure functions, Modifiers, Prototyping versus planning, Classes and methods: Object-oriented features, Printing objects, Another example, The `__init__` method, Operator overloading, Type-based dispatch, Polymorphism. Text book 2: Chapters 15

18

18

Number of Hours:9

Module-5

HTTP, The World

15

s simplest Web Browser, Retrieving an image over HTTP, Retrieving web pages with `urllib`, XML, Parsing XML, Looping through nodes, JSON, What is database?, Database Concepts, Database Browser, Creating a database table, SQL, Spidering Twitter, Basic data modeling, Programming with multiple tables, Three kinds of Keys, JOIN

15

Text book 3: Chapters12,13(Upto13.5),15

Number of Hours:8

Suggested Learning Resources: (Text Book/ Reference Book/ Manuals): Text books:

1.

Al Sweigart,

15

Automate the Boring Stuff with Python

16

,1st Edition, No Starch Press, 2015. (Available under CC-

BY

-

NC

-SA license at <https://automatetheboringstuff.com/>)

2.

Allen B Downey,

15

Think Python: How to Think Like a Computer Scientist

16

, 2nd Edition, Green Tea Press, 2015. (Available under CC-

BY

-NC license at <http://greenteapress.com/thinkpython2/thinkpython2.pdf>)

(Chapters 15 - 18) (Download pdf/html files from the ab

ove links)

3.

Charles R. Severance,

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Python for Everybody: Exploring Data Using Python 3

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, 1st, Create Space

Independent Publishing Platform, 2016

Web links and Video Lectures (e-Resources):

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https://www.youtube.com/watch?v=_xQNeOTRyig

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<https://www.youtube.com/watch?v=kqtD5dpn9C8>

Teaching-Learning Process (Innovative Delivery Methods): The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching-learning process and facilitate the achievement of course outcomes.

1.

Lecture method (L) does not mean only the traditional lecture method, but a different type of teaching

method may be adopted to develop the outcomes.

2.

Show Video/animation films to explain the functioning of various programming concepts.

3.

Adopt Problem Based Learning (PBL), which fosters students

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Analytical skills, develop thinking skills such as the ability to evaluate, generalize, and analyse information rather than simply recall it.

4.

Show the different ways to solve the same problem and encourage the students to come up with their own creative ways to solve them.

5.

Discuss how every concept can be applied to the real world - and when that's possible, it helps improve the students' understanding.